**9. Java Script Functions and Events**
**a. Design a appropriate function should be called to display**
**i. Factorial of that number**
**ii. Fibonacci series up to that number**
**iii. Prime numbers up to that number**
**iv. Is it palindrome or not**

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript Functions and Events</title>
  <script>
    function calculate() {
      let n = parseInt(document.getElementById("num").value);
      let output = "";

      // Factorial
      function factorial(n) {
        let fact = 1;
        for (let i = 1; i <= n; i++) {
          fact *= i;
        }
        return fact;
      }

      // Fibonacci
      function fibonacci(n) {
        let a = 0, b = 1;
        let series = "";
        while (a <= n) {
          series += a + " ";
          let temp = a + b;
          a = b;
          b = temp;
        }
        return series;
      }

      // Prime Numbers
      function primes(n) {
        let primeList = "";
        for (let i = 2; i <= n; i++) {
          let isPrime = true;
          for (let j = 2; j <= Math.sqrt(i); j++) {
            if (i % j == 0) {
              isPrime = false;
              break;
            }
          }
          if (isPrime) {
            primeList += i + " ";
```

```
                }
            }
            return primeList;
        }

        // Palindrome
        function isPalindrome(n) {
            let original = n.toString();
            let reversed = original.split("").reverse().join("");
            return original === reversed;
        }

        output += "<b>Factorial:</b> " + factorial(n) + "<br>";
        output += "<b>Fibonacci Series:</b> " + fibonacci(n) + "<br>";
        output += "<b>Prime Numbers:</b> " + primes(n) + "<br>";
        output += "<b>Palindrome:</b> " + (isPalindrome(n) ? "Yes" : "No");

        document.getElementById("result").innerHTML = output;
    }
  </script>
</head>
<body>
  <h2>JavaScript Functions and Events</h2>
  Enter a Number:
  <input type="number" id="num">
  <button onclick="calculate()">Submit</button>
  <h3>Result:</h3>
  <div id="result"></div>
</body>
</html>
```

**b. Design a HTML having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display**
**i. Factorial of that number**
**ii. Fibonacci series up to that number**
**iii. Prime numbers up to that number**
**iv. Is it palindrome or not**

```
<!DOCTYPE html>
<html>
<head>
 <title>JavaScript Functions and Events</title>
 <script type="text/javascript">
  // Utility: Reads the input value and converts it to an integer.
  function getInputNumber() {
   return parseInt(document.getElementById("numberInput").value, 10);
  }

  // Utility: Displays the provided message in the result div.
```

```
function displayResult(message) {
  document.getElementById("result").innerHTML = message;
}

// Function to calculate the factorial of the entered number.
function computeFactorial() {
  var num = getInputNumber();
  if (isNaN(num) || num < 0) {
    displayResult("Please enter a non-negative integer.");
    return;
  }
  var factorial = 1;
  for (var i = 2; i <= num; i++) {
    factorial *= i;
  }
  displayResult("Factorial of " + num + " is " + factorial + ".");
}

// Function to display the Fibonacci series up to the entered number.
function computeFibonacci() {
  var num = getInputNumber();
  if (isNaN(num) || num < 0) {
    displayResult("Please enter a non-negative integer.");
    return;
  }
  var fibSeries = [];
  var a = 0, b = 1;
  // Always include 0 in the series.
  if (num >= 0) {
    fibSeries.push(a);
  }
  // Include 1 if within the limit.
  if (num >= 1) {
    fibSeries.push(b);
  }
  var next = a + b;
  while (next <= num) {
    fibSeries.push(next);
    a = b;
    b = next;
    next = a + b;
  }
  displayResult("Fibonacci series up to " + num + ": " + fibSeries.join(", "));
}

// Function to list all prime numbers up to the entered number.
function computePrimes() {
  var num = getInputNumber();
  if (isNaN(num) || num < 2) {
    displayResult("Please enter an integer greater than or equal to 2.");
```

```
        return;
      }
      var primes = [];
      for (var i = 2; i <= num; i++) {
        var isPrime = true;
        // Check for factors up to the square root of i.
        for (var j = 2; j <= Math.sqrt(i); j++) {
          if (i % j === 0) {
            isPrime = false;
            break;
          }
        }
        if (isPrime) {
          primes.push(i);
        }
      }
      displayResult("Prime numbers up to " + num + ": " + primes.join(", "));
    }

    // Function to check whether the entered number is a palindrome.
    function checkPalindrome() {
      var input = document.getElementById("numberInput").value;
      if (input === "") {
        displayResult("Please enter a number.");
        return;
      }
      var reversed = input.split("").reverse().join("");
      if (input === reversed) {
        displayResult(input + " is a palindrome.");
      } else {
        displayResult(input + " is not a palindrome.");
      }
    }
  </script>
</head>
<body>
  <h1>JavaScript Functions and Events</h1>
  <p>
    <label for="numberInput">Enter a number:</label>
    <input type="text" id="numberInput" placeholder="Enter number" />
  </p>
  <p>
    <button onclick="computeFactorial()">Factorial</button>
    <button onclick="computeFibonacci()">Fibonacci</button>
    <button onclick="computePrimes()">Prime</button>
    <button onclick="checkPalindrome()">Palindrome</button>
  </p>
  <div id="result" style="margin-top:20px; font-size:1.2em;"></div>
</body>
</html>
```

**c. Write a program to validate the following fields in a registration page**
**i. Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)**
**ii. Mobile (only numbers and length 10 digits)**
**iii. E-mail (should contain format like xxxxxxx@xxxxxx.xxx)**

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form Validation</title>
  <style>
    .error { color: red; }
    .success { color: green; }
    body { font-family: Arial, sans-serif; }
    form div { margin-bottom: 10px; }
  </style>
  <script type="text/javascript">
    function validateForm(event) {
      // Prevent form submission (page refresh)
      event.preventDefault();

      // Clear previous messages
      document.getElementById("message").innerHTML = "";

      // Get field values and trim any extra spaces
      var name = document.getElementById("name").value.trim();
      var mobile = document.getElementById("mobile").value.trim();
      var email = document.getElementById("email").value.trim();

      // Array to hold error messages
      var errors = [];

      // Validate Name:
      // - Starts with an alphabet
      // - Followed by alphanumeric characters
      // - Total length not less than 6 characters
      var nameRegex = /^[A-Za-z][A-Za-z0-9]{5,}$/;
      if (!nameRegex.test(name)) {
        errors.push("Name must start with an alphabet, followed by alphanumeric characters,
and be at least 6 characters long.");
      }

      // Validate Mobile:
      // - Only digits
      // - Exactly 10 digits
      var mobileRegex = /^\d{10}$/;
      if (!mobileRegex.test(mobile)) {
        errors.push("Mobile number must be exactly 10 digits and contain only numbers.");
      }
```

```
      // Validate E-mail:
      // - Basic pattern: one or more allowed characters, an @, one or more allowed characters,
a dot, and at least 2 letters
      var emailRegex = /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;
      if (!emailRegex.test(email)) {
        errors.push("E-mail must be in the format xxxxxxx@xxxxxx.xxx.");
      }

      // Display messages
      var messageDiv = document.getElementById("message");
      if (errors.length > 0) {
        var errorHTML = "<ul>";
        errors.forEach(function(error) {
          errorHTML += "<li class='error'>" + error + "</li>";
        });
        errorHTML += "</ul>";
        messageDiv.innerHTML = errorHTML;
      } else {
        messageDiv.innerHTML = "<p class='success'>Registration Successful!</p>";
      }
    }
  </script>
</head>
<body>
  <h1>Registration Form</h1>
  <form onsubmit="validateForm(event)">
    <div>
      <label for="name">Name:</label>
      <input type="text" id="name" placeholder="Enter your name" required>
    </div>
    <div>
      <label for="mobile">Mobile:</label>
      <input type="text" id="mobile" placeholder="10-digit mobile number" required>
    </div>
    <div>
      <label for="email">E-mail:</label>
      <input type="text" id="email" placeholder="example@domain.com" required>
    </div>
    <button type="submit">Register</button>
  </form>
  <div id="message" style="margin-top:20px;"></div>
</body>
</html>
```